PCA applied to Galaxies Dataset

Corrado Navilli - S291251

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1 Introduction

The scope of this project is to study the dimensionality reduction of a 65-dimensional dataset concerning the estimation of the *redshift* of some Galaxies. By exploiting PCA, it is possible to reduce the number of features and therefore improve machine learning algorithms based on distance computations, such as *K-Nearest Neighbors*.
The homework has been developed using Pyhton.

2 Extraction of the Working Dataset

Exploiting the Python libraries **pandas** and **numpy**, the CSV file containing 3462 records is loaded and split into two subsets:

- Subset for PCA and KNN training (2500 rows)
- Evaluation set for KNN (962 rows)

The corresponding datasets are saved as CSV files.

2.1 Exploratory Data Analysis

By performing exploratory data analysis, it is possible to see that the training dataset requires some preprocessing.

- **Missing values**: There are some missing values in the columns *VbMag* (0.04
- **Columns selection**: Among the 65 original columns of the dataset, 1 is used as a target (*Mcz*) and the other 3 are excluded since they are related to it (*e.Mcz*, *MCzml*, *chi2red*). Another column (*Nr*) is removed from the dataset because it is only a numerical identifier of the records. The final number of features is 60.

- **Outliers removal**: In order to clean the data from possible mistakes (some datapoints had suspicious values, as could be seen from the scatter plot computed in the first version of the project included in the appendix), an outlier removal technique based on z-scores is applied. This involves applying standardization to the columns of interest and then removing those rows having at least a feature value greater than 3. The standardized version of the matrix is used for computing the PCA based on the correlation matrix in the second version of the project. This step reduced the number of datapoints to 2116 records.

Finally, the 60 feature columns' data can be standardized or not. In the **first version** of the project, the data were **not standardized**.

Given the original data matrix S, the standardization leads to $\hat{S}$, whose entries are defined with the formula above.

3 PCA

Since the data related to the features have not been standardized, the PCA will be performed on the Sample Covariance Matrix. Such a matrix is a square matrix defined as follows:

$$\mathrm{Cov}(S)_{ij} = \mathrm{Cov}(S_i, S_j) \quad i, j = 1, \ldots, N$$

On the main diagonal ($i = j$), all the entries are equal to $\sigma^2(S_i)$.

Instead of computing one by one the entries of the Sample Covariance Matrix, it is possible to obtain it with a matrix multiplication based on the centered sample matrix $\bar{S}$ as follows:

$$\mathrm{Cov}(S) = \frac{1}{N-1}\bar{S}^T\bar{S}$$

where N is the number of samples.

Once the Sample Covariance matrix is available, it is possible to compute its eigendecomposition as:

$$P\Lambda P^T = \mathrm{Cov}(S)$$

P is the matrix whose columns describe the directions of maximal variance of the data, and Λ is instead a diagonal matrix containing all the eigenvalues of the covariance matrix in decreasing order. Such eigenvalues represent the amount of variance along the direction of the corresponding eigenvectors (i.e., columns of P). By dividing each eigenvalue by the sum of all the eigenvalues, it is possible to retrieve the ratio of the explained variance associated with that specific component.

In the project, the PCA decomposition has been performed exploiting the PCA class of the `sklearn` package.

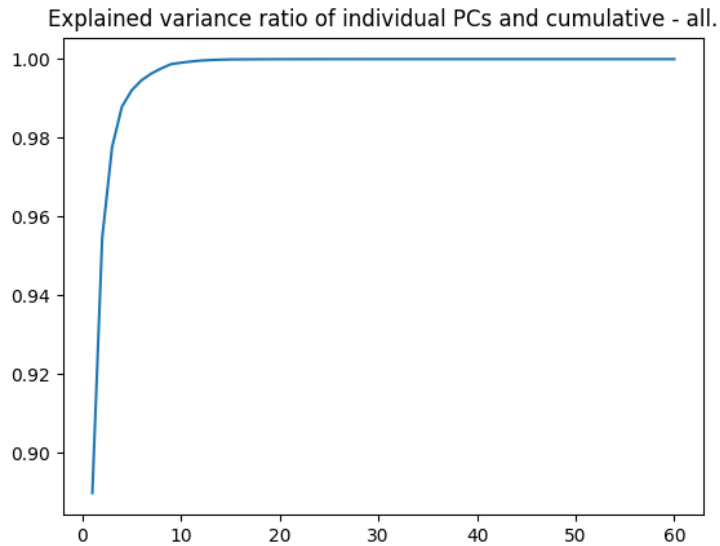

Figure 1: PCA cumulative explained variance ratio.

1. Sample data are provided as input to PCA without specifying the number of components.

2. The cumulative sum of explained variance ratios is computed to get a first insight into the most relevant components.

The results of this first analysis are displayed in the graph in Figure 1. It can be seen that the first 20 components explain almost 100% of the variance, and the first 10 more than 99%. Between 10 and 20, the line shows a plateau. Furthermore, the first component seems to explain almost 65% of it.

For these reasons, the focus was set on the first 10 components, resulting in the graph in Figure 2 displaying individual explained ratios as bar plots and the cumulative sum as a line chart:

2 components are enough to achieve the target of 95%, while for 99% the minimum number is 5. Furthermore, it can be seen that already from the 4th component the individual contribution is minimal (2.3%).

For all these reasons, the **number of components** was then fixed to **5** (and the PCA was recomputed providing this value as an argument, in order to simplify the coding operations).

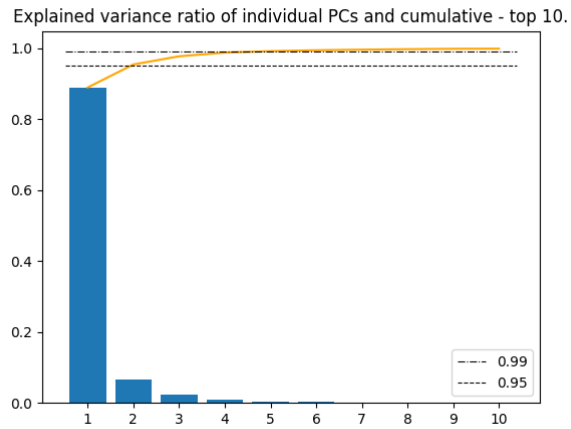

Figure 2: Explained Variance Ratios

4 Interpretation of the components

Since PCA outputs a series of linear recombinations of the original features, the principal components themselves are used as new features. As a consequence, the explicit interpretability of the feature meaning is lost. This can be somehow fixed.

Another goal of PCA, from a statistical perspective, is to detect underlying correlations between the original features by examining the entries of the principal components. This enables the renaming of the components, providing a level of interpretability (for which a good level of domain knowledge is required).

Since there are 60 original features, it is challenging to provide a global interpretation considering all the entries together. Therefore, after an empirical evaluation, it has been chosen to consider only the entries with an absolute value greater than or equal to **0.10** (except for the 3rd component, for which the threshold is **0.15**, as with 0.10, the entries are still too numerous). This allows us to limit the number of significant entries.

In the next sections, the vectors referring to the first 5 components (the ones explaining more than 99% of the variance in the sample distribution) will be displayed and explained.

4.1 1st component

In this component, the most relevant features are those referring to the absolute magnitude of the galaxy measured according to different bands. These magnitudes are not independent of each other; therefore, the redundant information may be the cause of the bars having almost the same length. The important thing is that they are indeed all considered together, as an expected output of a

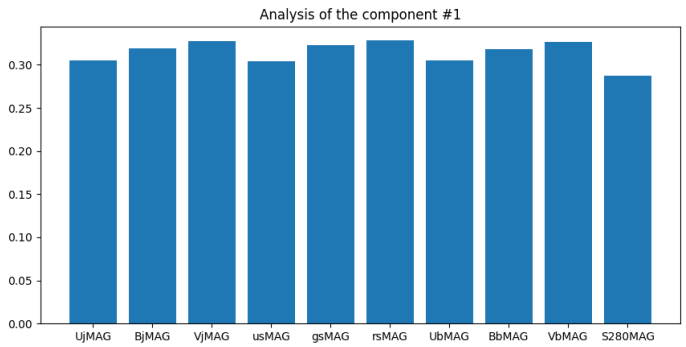

Figure 3: Interpretation of 1st component

PCA operation: correlated features are grouped together. Since the magnitude represents the luminosity of a galaxy, we could rename this first component as "**Luminosity of the Galaxy**". [Figure 3]

4.2 2nd component

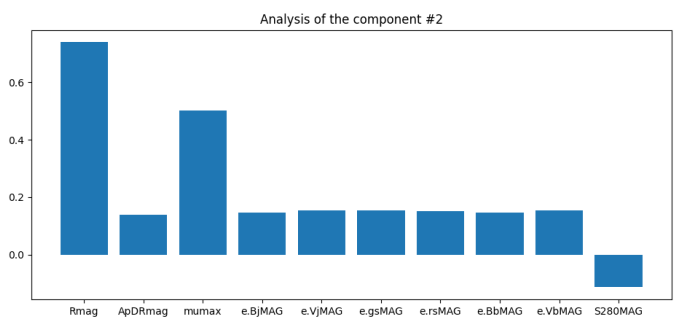

Figure 4: Interpretation of 2nd component

Rmag and *mumax* are the entries with the highest values. *Mumax*, in particular, is the central surface brightness of the object in the R band, while *Rmag* is the total red band magnitude, and their difference could be an indicator of galaxy size. This principal component also takes into account the relative error measurements on magnitudes, although at a lower level. That being said, we could refer to this component as "**Size of the Galaxy (and Measurement Errors)**". [Figure 4]

4.3 3rd component

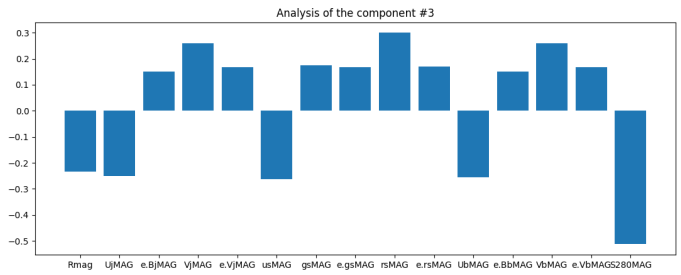

Figure 5: Interpretation of 3rd component

The third component is more complex, but again the main entries are those referring to magnitude measures. In particular, the negative values refer to *Rmag*, *Ujmag*, *usmag*, *ubmag*, and *S280*. Features starting with "u" refer to wavelengths belonging to the ultraviolet band, while *Rmag* is in the red band. The meaning of *S280* is not completely clear; it represents a different magnitude measurement system, but the details are not provided.

The main positive values are instead *VjMAG*, *gsMAG*, *rsMAG*, *VbMAG*, where measures starting with "V" refer to visible light, "g" represents green light, and "r" again denotes red light but considering the SDSS system.

It may seem that on one side, we have visible band light, and on the other side, the bands above and below it. Furthermore, given the lack of information on *S280*, we could exploit the insights to guess a correlation of it with Ultraviolet/Infrared bands. With acknowledgment of the lack of domain expertise, we could refer to this principal component as "**Magnitude in the S280 system and Ultraviolet band Versus Magnitude in Visible Light**". [Figure 5]

4.4 4th component

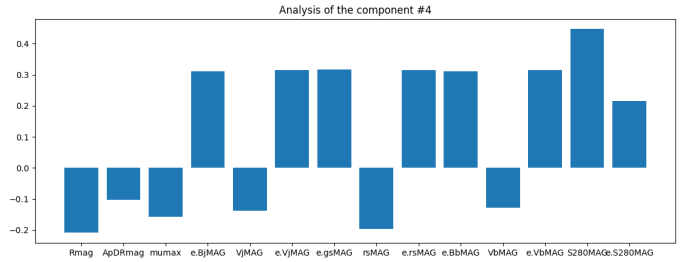

Figure 6: Interpretation of 4th component

In this case, the predominant values are positive and refer to the measurement error on the green, blue, and violet bands, with the unknown *S280* value outranging all the others. On the negative side, but with shorter length, we have measures in the red band. Without repeating similar considerations made for the 3rd component, we could refer to this as "**Error on Blue, Green, Violet Band Measures vs. Measures on Red Band**". [Figure 6]

4.5 5th component

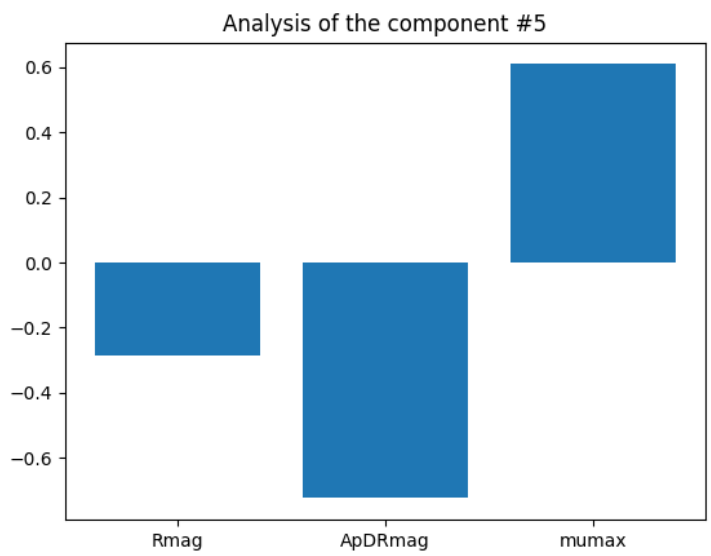

Figure 7: Interpretation of 5th component

The fifth component is all about the size of the galaxy. Same as explained for PC 2, but in this case, values are higher, and the contribution of error measurements is not included (given a threshold of 0.1). In particular, we can see that on the negative side, we have the features referring to the magnitude computed in the R band, while on the positive side, we have the measure of brightness. Therefore, also to distinguish from PC 2, we could interpret this as "**Size of the Galaxy Computed as Brightness vs. Magnitude, in R Band**". [Figure 7]

5 PCA Graphical representation

The application of PCA with 5 components allows us to project the original sample data onto a 5-dimensional space. This significantly reduces the number

of dimensions, but it is still not possible to visualize the data in their original space.

For this reason, the transformed data are plotted onto 2D planes and colored with hues related to the value of the target variable, *Mcz*.

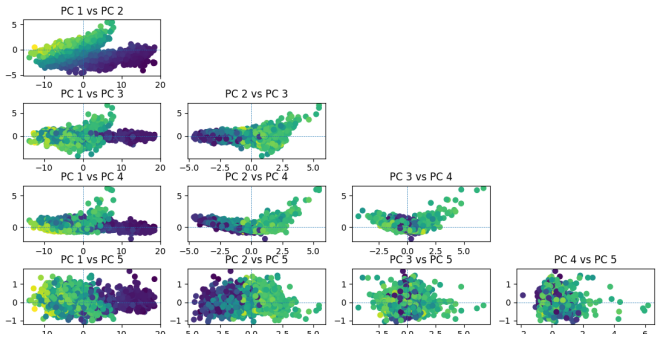

Figure 8: 2D representation of data, according to different pairs of the first PCs

This matrix of plots displayed in Figure 8 enables a first qualitative analysis of the discriminant effects of PCA. Although the linear separation of data is not one of the goals of PCA, it can be seen that points belonging to the same area of the plane have similar coloration. For example, by looking at the first column, we can see the plots referring to PC1 vs. all the others considered: blue points (those with lower values of *Mcz*, that is to say the redshift effect) are mainly in the bottom-right part of the plots, while the datapoints referring to galaxies with a higher redshift effect are on the left.

Another interesting observation is that as we move towards the left, the "columns" show more mixed datapoints, suggesting that moving to "less meaningful" principal components results in less discriminative power when considering only those as features.

As a last representation, 1D, 2D, and 3D plots are used to display the discriminative power of using the first 1, 2, and three principal components, respectively (the second plot is a repetition of what was already seen in the previous image).

6 K-Nearest Neighbors

In the last part of the pipeline, the goal is to approximate the value of *Mcz* for the test dataset (a regression task) using only the first 9 principal components. To achieve this, the model *KNeighborsRegressor* from the scikit-learn library is trained on the training data, which has been projected onto a 5-dimensional space. No fine-tuning is performed on the model, and the number of neighbors

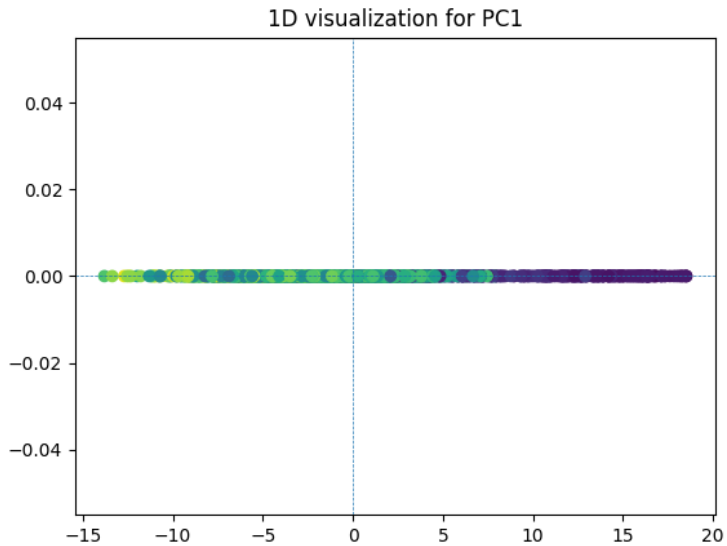

Figure 9: Projection of data onto first PC (1D).

is set to 5. Subsequently, the PCA transformation is applied to the test data, which are then fed to the model to obtain predictions.

The quality of the approximation is finally computed with respect to two metrics:

$$\text{Mean Absolute Error (MAE)} = \frac{1}{N}\sum_{i=1}^{N} |Mcz_i - \hat{Mcz}_i|$$

and

$$\text{Mean Relative Error (MRE)} = \frac{1}{N}\sum_{i=1}^{N} \left|\frac{Mcz_i - \hat{Mcz}_i}{Mcz_i}\right|$$

where N represents the number of data points in the test dataset, Mcz_i is the actual value of Mcz for the i-th data point, and $\hat{Mcz}_i$ is the predicted value.

The results are compared with the ones obtained without applying PCA to the original data, therefore performing calculations in 60 dimensions:

- With PCA (5 components):
  - MAE: 0.032833
  - MRE: 0.067298
- Without dimensionality reduction:

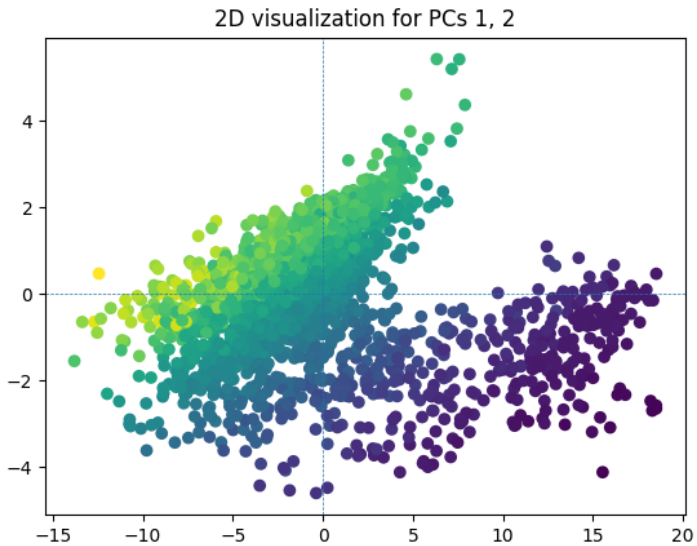

Figure 10: Projection of data onto the first 2 Pcs (2D).

- MAE: 0.033605
- MRE: 0.069371

It can be seen that both error measures improved, even though we used only a total of 5 features instead of 60 for modeling the data.

7 Further experiments: Standardization and Outliers removal

As a very last step of the project, a comparison was made by performing standardization of the data before applying PCA. Such a standardized version of the dataset has already been computed for the outliers removal step.

The choice of standardizing the data is related to the fact that different scales in the data may lead to an imbalance in the distance computation towards the features with a larger scale (and an algorithm such as KNN is distance-based).

The following formula summarizes the standardization operation (performed with the *StandardScaler* class of the **scikit-learn** library):

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where i is the row index and j is the column index.

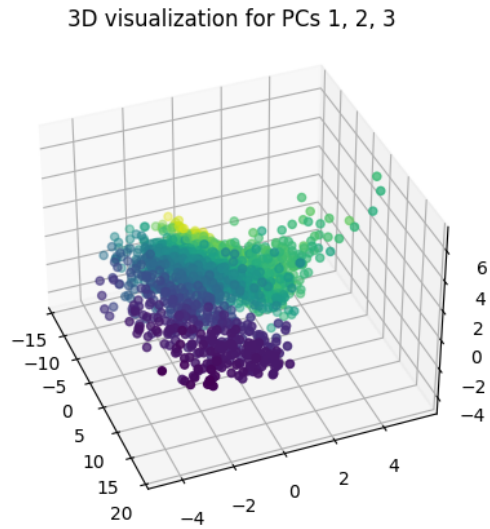

Figure 11: Projection of data onto the first 3 Pcs (3D).

Then, by using the standardized data, denoted as $\hat{S}$, it is possible to compute the **correlation matrix** as follows, without the need to calculate entry by entry:

$$\text{Corr}(S) = \frac{1}{N-1}\hat{S}^T\hat{S}$$

Results

The first thing to notice is that the number of components needed to achieve 99% of explained variability increased to 30. Then, by looking at just the first 4 components' values, it can be seen that features are differently recombined inside each component compared to PCA performed on the original data. This confirms the fact that although it is possible to perform PCA either on the original data or standardized ones (i.e., with the mean removal and division by standard deviation), the results may differ.

Finally, MAE and MRE were computed as well. In this case, though, the results were significantly worse than the approximation done without dimensionality reduction:

- PCA on standardized data (30 components):
  - MAE: 0.074828
  - MRE: 0.162650
- Without dimensionality reduction on standardized data:

- MAE: 0.074592
- MRE: 0.162237

We can see that the performance worsened in using the standardized version of data.